Stephen Kyle V. Laroco

Cainta, Rizal | skvlaroco@gmail.com | 09288703664/09523919333
https://www.linkedin.com/in/skvlaroco/ | https://laroco-datascience.netlify.app/

EDUCATION

Technological Institute of the Philippines - Manila **Quiapo, Manila**
4th Year in Bachelor of Science in Data Science and Analytics *October 2026*

- **Relevant Coursework:** R Programming, Quantitative Introductory Statistics for Data Science, Advanced Data Analytics, Data Management, Data Visualization, Data Structures and Algorithm, Statistics and Probability, Data Acquisition/Engineering, Data Mining and Warehousing, Information Management, Applied Statistics Linear Algebra with MATLAB, Machine Learning, Big Data Analytics, Deep Learning, Business Forecasting with Time-Series Analysis, Machine Learning for Business, Machine Learning Algorithms and Statistical Methods for Data Security, Complexity Analysis, Applications Development and Emerging Technologies, Operations Research/Optimization
- **GWA:** 1.95

LEADERSHIP EXPERIENCE & ACTIVITIES

Collaborative Analytics and Data Science Society (C.A.D.S.S) **Manila, Philippines**
Founding President *August 2025 - Present*

- Founded and led a student organization focused on analytics and data science, organizing workshops, training sessions, and collaborative projects
- Spearheaded initiatives to promote data literacy and technical skills among students through hands-on learning and events
- Managed organizational planning, partnerships, and team coordination to ensure successful execution of activities

Globe Digital Thumbprint Program (DTP) **Manila, Philippines**
Digital Thumbprint Program Ambassador *September 2025 – Present*

- Selected as one of the first Digital Thumbprint Program Ambassadors in the Philippines, supporting Globe's initiatives in promoting responsible and meaningful digital citizenship among Filipino youth.
- Participated in and helped promote digital literacy initiatives focused on responsible technology use, online safety, and positive digital engagement.

Microsoft **Manila, Philippines**
Alpha Microsoft Learn Student Ambassador *April 2024 – November 2024*

- Selected as an **Alpha Microsoft Learn Student Ambassador**, demonstrating leadership and commitment to the program.
- Mentored and supported fellow Student Ambassadors, both on and off-campus, to drive meaningful change.

College of Computer Studies Department Student Council **Manila, Philippines**
Treasurer *September 2024 - May 2025*

- Manages the council's financial resources, ensuring proper budgeting and allocation of funds.
- Assists the Governor in coordinating decision-making processes for smooth council operations.
- Maintains financial records and prepares reports to ensure transparency and accountability.

Amazon Web Services – Cloud Clubs T.I.P Manila **Manila, Philippines**
Lead Data Analyst *February 2024 – June 2026*

- Led the organization in planning and executing various tech events, workshops, and community initiatives, with the goal of promoting our mission to unite, equip, and empower students through technology.

Information and Computing Organization of Networked Students (I.C.O.N.S) **Manila, Philippines**
Vice President *August 2025 - June 2026*

- Assisted in leading organizational operations, supporting the execution of technical events and student engagement initiatives
- Collaborated with executive members to plan programs that enhance students' knowledge in computing and technology
- Helped oversee internal coordination, ensuring smooth communication and efficient project implementation

SKILLS & INTERESTS

Computer: WEKA, Python, R Programming, SQL, Excel, Power BI, Notion, Adobe Photoshop, Canva
Languages: Filipino (Native), English (Proficient)
Interests: Leadership, Personal Growth, Notion Productivity, Music, Gaming, Sports

PROJECTS

Machine Learning Analysis of Institutional Facilities and Student Motivation

- Developed and compared supervised learning models (**Decision Tree, Random Forest, Naive Bayes**) and unsupervised models (**K-Means Clustering**) to predict student motivation based on facility satisfaction.
- Achieved **100% classification accuracy** using Random Forest and Decision Tree models, demonstrating a highly structured relationship between physical/digital learning environments and student engagement.
- Applied **Principal Component Analysis (PCA)** for dimensionality reduction and 2D/3D visualization of motivational clusters.
- Utilized **Python** and advanced libraries to provide data-driven insights for institutional policy development.

Factors Affecting Student Academic Performance: A Statistical Study

- Conducted a quantitative research study using **computational statistics** and **Exploratory Data Analysis (EDA)** to identify significant factors influencing exam scores.
- Implemented **Multiple Linear Regression, T-tests, and ANOVA** to quantify the impact of demographic and socioeconomic variables such as parental education and test preparation.
- Engineered features (e.g., total_score) and performed rigorous data cleaning on a dataset of 1,000 observations to ensure analytical accuracy.
- Presented findings through a structured **data storytelling framework** using advanced visualizations like correlation heatmaps and boxplots.

Global Retail Performance & Profitability Analysis

- Analyzed a dataset of **51,000+ records** to identify underperforming business segments and optimize global sales efficiency.
- Developed an interactive dashboard visualizing **Sum of Profit by Country and Region** to support strategic decision-making.
- Evaluated contributing factors to financial outcomes, including pricing strategies, discount impacts, and shipping costs.

Customer Segmentation for High-End Product Targeting

- Applied data mining and warehousing techniques to the **AdventureWorksDW2019** database to identify customer segments likely to purchase premium products.
- Engineered time-based features and performed data reduction via **SQL queries** to streamline temporal analysis.
- Utilized **Python** for data transformation and **Power BI** for evidence-based visualization of regional and customer-specific sales trends.

School Resource Management System (SRMS)

- Designed and developed a transaction-based system in **Java** to automate the tracking and allocation of school department faculty resources.
- Applied **Object-Oriented Programming (OOP)** and complex **Data Structures** to implement real-time inventory management, transaction logging, and role-based access control.
- Created a **Graphical User Interface (GUI)** using Swing/AWT to facilitate intuitive check-in/check-out operations and automated data reporting

Business Forecasting with Time-Series Analysis for DepEd SHS Enrollment and Resource Planning

- Developed a time-series forecasting model using **Facebook Prophet** to predict 3-year SHS enrollment trends, achieving ≥80% accuracy (MAPE) on historical NCR data.
- Engineered a data-driven pipeline to compute classroom and teacher requirements from forecasts, translating predictions into actionable resource allocation insights.
- Built a PHP/Python-based analytics system with data validation, interactive visualization, and automated reporting (CSV/PDF) for decision support.

CAPSTONE

HR Chatbot for Employee Support

Client: *Information Technology Group, Inc.* *July 2026 - Present*

- Developing an intelligent HR chatbot to assist employees with HR-related concerns.
- Applying **NLP and Retrieval-Augmented Generation (RAG)** to provide relevant responses based on company HR resources.
- Developing the system using **Python, FastAPI, PostgreSQL, and local LLM technologies**.
- Collaborating with a company client to identify requirements and develop a practical solution.

HACKATHONS AND COMPETITIONS

Data Analyst, Champion Team – Data & AI Inclusion Hackathon (Grundfos x TIP)

February 2026

- Performed data cleaning, transformation, and exploratory analysis to extract actionable insights
- Supported the development of a data-driven solution focused on empowering local students
- Contributed to the team's championship win through analytical insights and data storytelling

Data Analyst, National Finalist (Shortlisted) – ASEAN Data Science Explorers 2026, Philippines

August 2026 - Present

- Built a Climate-Urban Vulnerability Index comparing five Philippine cities, combining official PAGASA, PSA, and NDRRMC data into a Hazard-Exposure-Poverty Gap framework aligned with SDG 11 and SDG 13
- Applied min-max normalization and equal-weighted scoring, and ran sensitivity analysis on alternative weighting schemes to test whether the city rankings held up
- Developed a what-if simulator and an interactive dashboard (Ligtas resident app and Command Center) to help LGUs prioritize resilience investment
- Presented findings through data storytelling, with transparent disclosure of data limitations and methodology